

EDUCATION

University of Prince Edward Island

Bachelor of Science in Computer Science, Minor in Economics

Expected May 2027

- Data Structures and Algorithms, Software Engineering, Operating Systems, System Design, Machine Learning.

EXPERIENCE

Software Engineer Intern

Mackenzie Investments (IGM Financial)

May 2026 – Aug 2026

- Built a multi-solution analytics hub for the Product Department, using Flask, React, TypeScript, and Google BigQuery, giving investment teams one place to explore fund data instead of scattered reports.
- Built two analytics modules: Fund Health, which tracks fund performance and hit rates, and Brinson Attribution, which decomposes benchmark-relative returns into allocation and selection effects.
- Managed all frontend state through Redux Toolkit and RTK Query, keeping filters, cached data, and UI state in sync across both solutions.
- Built the data layer with SQLAlchemy and Marshmallow on a 23-table BigQuery warehouse, serving live data for 91 funds.
- Secured the platform with HashiCorp Vault for credentials and Microsoft Entra ID single sign-on with role-based access, backed by 89 automated tests.
- Automated artifact publishing and security scanning using GitHub Actions and JFrog Artifactory and collaborated with the infrastructure team on GCP deployment.

Teaching Assistant

University of Prince Edward Island

Sep 2025 – Apr 2026

- Mentored 40+ students in Java, Python, and C++ programming labs, guiding them through debugging, data structures, and algorithmic problem-solving.

Software Engineer Intern

British Columbia Electronic Library Network

Jan 2025 – Apr 2025

- Improved system interoperability for digital repositories by integrating external REST APIs and mapping metadata schemas, enabling automated retrieval of Traditional Knowledge (TK) Labels across Islandora-based platforms.
- Resolved production issues and enhanced platform reliability by debugging backend services, analyzing logs, and implementing targeted code fixes within a Drupal-based web application.

PROJECTS

MapleNest | Node.js • Express • Supabase • React • Tailwind CSS

- Built a full-stack housing marketplace enabling students to search and filter rental listings by location, price, and property type through RESTful APIs.
- Designed modular Express services and reusable responsive React components for desktop and mobile devices.
- Implemented Supabase authentication and database access controls to protect user accounts, rental listings, and application data.

Monopoly Strategy Simulation | Java

- Developed an extensible Java simulation engine for modeling financial transactions and asset-allocation strategies using modular object-oriented components.
- Implemented four algorithmic AI strategies and conducted 60+ simulations, generating comparative metrics to evaluate decision-making performance.

ASHE: Lords of the Orun | Unity, C#

- Designing and building a 2D pixel-art action RPG in Unity (C#) rooted in Yoruba mythology, featuring three elemental combat paths and a hidden morality system that shapes boss encounters.
- Building a five-room playable vertical slice with a complete combat kit, boss encounter, and morality-based decision branch to validate the core gameplay loop.

TECHNICAL SKILLS

- Languages: Java, Python, C++, C#, JavaScript, TypeScript, SQL, HTML/CSS
- Frameworks & Libraries: React, Next.js, Node.js, Express, Flask, Redux Toolkit, Spring Boot, SQLAlchemy, Marshmallow
- Databases & Cloud: Google BigQuery, PostgreSQL, MongoDB, Redis, Supabase, AWS, Google Cloud Platform
- Tools & Infrastructure: Git, GitHub Actions, Docker, Linux, JFrog Artifactory, HashiCorp Vault, Microsoft Entra ID, OAuth 2.0/OIDC, Postman, JUnit, Unity, Drupal

CERTIFICATIONS

- Google Cloud Fundamentals